Advanced high school · geometry

Rotations and complex numbers

Progressive exercises

Exercise 1

Find $i(2-3i)$, then its modulus.

Exercise 2

Find the successive images of $1+i$ under four multiplications by i.

Exercise 3

What is the centre of $z' = iz + 2$?

Exercise 4

For $z' = a + i(z-a)$, prove $|z'-a| = |z-a|$.